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The Hidden Cost of the Cloud

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Executive Summary

Data centers underpin AI training, cloud computing, and streaming, but their cooling systems consume large volumes of freshwater. This report uses Google's data center in The Dalles, Oregon as a case study: a thirteen-month public-records dispute revealed that facility water use grew from 394 million liters in 2010 to an estimated 2.1 billion liters by 2025 - about 40% of the drought-affected city's supply (Rogoway, 2022; Boudreau, 2026). The core problem is information asymmetry: host communities supply a scarce resource without being able to verify its scale, violating informed consent and raising environmental-justice concerns. This report proposes mandatory, third-party-verified, facility-level water reporting - a policy and technical framework that closes this gap without dictating how much water a facility may use - and argues that disclosure, not engineering alone, is the most immediately actionable response given that U.S. data center electricity demand is projected to double by 2028 (Shehabi et al., 2024).

1.0 Introduction

Data centers now perform trillions of computations per second for AI training, cloud computing, and streaming, but this growth has a hidden physical cost: land, electricity, and large volumes of freshwater used for evaporative cooling. As computing capabilities scaled rapidly over the past two decades, operators maximized output by improving chip efficiency and by redesigning cooling systems around evaporative towers, which circulate municipal water to absorb heat before a portion evaporates and is permanently lost. Between 2010 and 2018, global computing demand grew more than sixfold while electricity per unit of computation fell to a quarter of its prior level (Masanet et al., 2020), yet operators were slow to measure and disclose the water this cooling required.

The demand for computing power has since accelerated sharply with the rise of AI. U.S. data center electricity consumption rose, with demand more than doubling between 2017 and 2023 (Shehabi et al., 2024). Google's own environmental disclosures indicate its 2025 potable water use exceeded 41 billion liters, more than double its 2021 figure (Google, 2026), and nationally, total data center water consumption surpassed 860 billion liters in 2023 (Shehabi et al., 2024). This report asks how water consumption by data centers should be governed so that host communities can make informed decisions about the facilities they host? Section 2 examines Google's Dalles, Oregon facility as a case study. Section 3 identifies the underlying problem of information asymmetry. Section 4 presents the ethical principles at stake. Section 5 proposes a mandatory disclosure framework. Section 6 concludes.

2.0 Case Study: Google's Data Center in The Dalles, Oregon

2.1 The Disclosure Dispute

The Dalles, a city of roughly 16,000 residents on the Columbia River, has hosted a Google data center campus since 2006, drawn by inexpensive hydroelectric power, fiber connectivity, and local tax incentives. For over a decade the scale of the facility's water use was not public. When a reporter for The Oregonian sought consumption records under Oregon's public records law, the city initially withheld them, and Google - whose contract classified the figures as a trade secret - funded the city's legal effort to block release, despite the county District Attorney having already ordered disclosure. The resulting lawsuit delayed release by thirteen months; the city abandoned the case in December 2022 and released ten years of data (Rogoway, 2022).

The released records showed annual consumption grew from 394 million liters in 2010 to 1.3 billion liters in 2021, more than a threefold increase over eleven years. Subsequent reporting indicates it reached approximately 2.1 billion liters by 2025, close to 40% of the city's total water supply during sustained regional drought (Boudreau, 2026).

2.2 A Representative Case

The Dalles is illustrative rather than exceptional. Communities hosting data centers in Georgia, Arizona, and Virginia have raised similar concerns about opaque water and power allocations, sometimes only learning the scale of a facility's resource draw after construction was

already underway (Harvard Gazette Staff, 2026), and Shah and Hall (2026) identify facility-level opacity as a recurring driver of water-insecurity concern nationwide. Notably, Google's 2026 environmental report acknowledges the reporting gap this case exposed, describing efforts to make water use less of a corporate "black box" (Boudreau, 2026), a useful signal that facility-level disclosure is achievable.

3.0 Problem: Information Asymmetry in Water Disclosure

No federal, state, or provincial law currently requires facility-level water disclosure; what exists is voluntary and aggregated at the corporate level. In The Dalles, residents learned the scale of local water use only after a thirteen-month legal fight (Rogoway, 2022); in jurisdictions without an equivalent public-records mechanism, or where contracts classify the figures as trade secrets, residents may have no path to this information at all (Shah & Hall, 2026).

This opacity causes real harm: it prevents municipalities from weighing data center hosting against agricultural and residential water needs before contracts are signed; it concentrates environmental risk in smaller, often rural communities with limited negotiating leverage (Harvard Gazette Staff, 2026); and it obscures, behind global aggregates such as Google's reported 41 billion liters (Google, 2026), which specific watersheds bear the cost. Crucially, this is a governance failure, not a technical one. Zhao et al. (2021) show operators already run automated systems to regulate power consumption in near real time, infrastructure that could be extended to meter water with comparable precision.

4.0 Ethical Principles and Social Danger at Stake

4.1 Informed Consent

A community that supplies a resource to a third party has a reasonable claim to know, in advance and on an ongoing basis, the scale and impact of that supply. In The Dalles, residents effectively subsidized a decade of cooling with municipal water before they were permitted to know how much water was involved, let alone consent to that scale of use (Rogoway, 2022). Consent obtained only after litigation is not meaningful consent.

4.2 Distribution of Environmental Cost

The benefits of data centers (AI capability, cloud services, streaming) appeal to a broad, often global, user base and to corporate shareholders, while the water and land costs are absorbed locally, frequently by communities with limited political leverage (Shah & Hall, 2026; Harvard Gazette Staff, 2026). This asymmetry between diffuse benefit and concentrated cost is a recurring pattern in environmental justice, and the data center sector reproduces it closely: the roughly 40% share of The Dalles's water supply reportedly consumed by a single campus (Boudreau, 2026) is a burden felt only by that city's residents, not by the millions of remote users whose AI queries or streamed video the facility supports.

4.3 Limits of Trade-Secret Protection

Trade-secret protection exists to safeguard genuinely competitive information, not aggregate consumption figures. Using confidentiality law to withhold environmentally material data, as occurred in The Dalles, overextends a protection intended for a narrower purpose and sets a troubling precedent for other opaque infrastructure.

4.4 Social Danger

Left unaddressed, this pattern risks normalizing information asymmetry as the default relationship between operators and host communities, increasing the odds that water-stressed regions commit supply without adequate foresight, and eroding public trust in both companies and local governments. These risks compound as U.S. data center electricity demand is projected to double by 2028 (Shehabi et al., 2024), extending this asymmetry to many more communities before any governance mechanism exists.

5.0 Proposed Solution: Mandatory, Verified Facility-Level Water Reporting

5.1 What and Why

This report proposes a legal requirement, implemented at the state or provincial level and ideally harmonized nationally, that data center operators above a size threshold publicly report, quarterly, each facility's water withdrawal and consumption, its water source, and its Water Usage Effectiveness (WUE, liters/kWh), on a regulator-hosted public dashboard subject to independent audit. The threshold must hold for the whole site instead of individual buildings such that the criteria holds up for the entirety of the campus and not be subdivided amongst several buildings which can report different levels of consumption. This targets the root cause

rather than only its symptom: it does not mandate reduced consumption, but it is a precondition for any informed local decision-making.

5.2 Administration and Enforcement

For change to be installed there must be an enforcement mechanism. Administrative authority should fall on a regulatory organization who already has responsibility over water permission. They must uphold the regulatory standards to authorize the usage of water. Should firms conceal their findings, this governing body can refute the right to access municipal water. Financial penalties should reinforce late reporting or skewed results.

Auditing must be independent. Due to conflict of interest and financial incentives, operators do not have the right to selectively choose which bodies verify their findings. Enforcement by regulatory bodies and the incorporation of independent auditing can delegate the burden from the public to the industries generating the reporting.

5.3 Target Audience

This proposal is addressed to bodies who govern over water legislation. Secondary to the state, is the municipal councils who currently negotiate water agreements without access to the quantitative data it governs. The Dalles made decisions unaware of the true figures it was to affect. Lastly, this proposal is aimed at the federal level, where full disclosure can converge into proper metrics.

5.4 Technical Feasibility

Reliable disclosure depends on metering at the cooling-loop level, distinct from existing plant-wide utility billing. Zhao et al. (2021) show operators already deploy automated, granular monitoring to regulate power consumption in near real time; extending comparable metering to water, and reporting through a standardized API, is a modest technical extension of infrastructure that already exists, minimizing the administrative burden of quarterly disclosure and reducing the risk of manual under-reporting.

5.5 Complementary Engineering Alternatives

Disclosure can be paired with engineering approaches that reduce freshwater draw directly: closed-loop and liquid-immersion cooling reduce evaporative loss compared to open

evaporative towers, and siting facilities to draw on reclaimed wastewater or seawater rather than potable supply is another option. Highlander's underwater deployments in Hainan and Shanghai use ambient seawater for passive cooling, eliminating freshwater withdrawal entirely (Datacenter Dynamics, 2026). More speculative still, proposals for space-based data centers would remove terrestrial water and land competition altogether, though such designs remain in early conceptual and economic stages and face significant launch-cost and reliability barriers before they could meaningfully offset near-term terrestrial demand (Forethought, 2026; SemiAnalysis, 2026). These alternatives are best understood as a longer-horizon complement to disclosure, not a substitute for it: they address consumption directly but do nothing, on their own, to resolve the informed-consent problem in facilities that continue to use conventional cooling in the interim.

6.0 Conclusion

The Dalles case shows a structural pattern: the AI-driven expansion of data centers has outpaced the governance needed to make its resource costs visible to host communities. A thirteen-month lawsuit was required merely to learn that a single facility's water use had grown more than fivefold in a decade (Rogoway, 2022; Boudreau, 2026). The problem is not just consumption, but an absence of informed consent, and the proposed solution: mandatory, verified, facility-level reporting, directly closes that gap using metering infrastructure that already exists.

Its limits should be stated plainly: disclosure does not itself reduce consumption, does not address facilities already built without such requirements, and depends on political will to legislate a harmonized standard rather than a fragmented patchwork. Complementary engineering approaches remain part of a fuller response. As U.S. data center demand is projected to double or triple by 2028 (Shehabi et al., 2024), mandatory, community-visible water reporting should be treated as a baseline governance requirement for the next generation of data center infrastructure, not an optional best practice.

References

- Bloomberg. (2025). AI data centers are straining global power grids and water supplies.
<https://www.bloomberg.com/news/articles/2025-ai-data-center-energy-water>
- Boudreau, C. (2026). Data center water use can be a "black box." Google aims to change that.
Latitude Media.
<https://www.latitudemedia.com/news/data-center-water-use-black-box-google-trying-to-change/>
- Datacenter Dynamics. (2026). Highlander advances commercial underwater data center deployments in Hainan and Shanghai. DCD.
<https://www.datacenterdynamics.com/en/news/highlander-underwater-data-center-hainan-shanghai/>
- Forethought. (2026). Will we really put data centers in space? Forethought Blog.
<https://www.forethought.ai/blog/will-we-really-put-data-centers-in-space>
- Google. (2026). Google environmental report 2026. Google Sustainability.
<https://sustainability.google/reports/environmental-report-2026/>
- Government of Canada. (2026). AI for all strategy. Innovation, Science and Economic Development Canada.
<https://www.canada.ca/en/innovation-science-economic-development/news/2026/ai-for-all-strategy.html>
- Harvard Gazette Staff. (2026, April). Why are communities pushing back against data centers?
The Harvard Gazette.
<https://news.harvard.edu/gazette/story/2026/04/why-are-communities-pushing-back-against-data-centers/>
- Masanet, E., Shehabi, A., Lei, N., Smith, S., & Koomey, J. (2020). Recalibrating global data center energy-use estimates. *Science*, 367(6481), 984-986.
<https://doi.org/10.1126/science.aba3758>
- Rogoway, M. (2022, December 16). Google's water use doubles in The Dalles, city records show.
The Oregonian.

<https://www.oregonlive.com/silicon-forest/2022/12/googles-water-use-doubles-in-the-dalles-city-records-show.html>

SemiAnalysis. (2026). To boldly go: The case for space datacenters.

<https://www.semianalysis.com/p/to-boldly-go-the-case-for-space-datacenters>

Shah, S. H., & Hall, C. (2026). Four water insecurity concerns about datacenters driving the AI revolution. *PLOS Water*, 5(1), Article e0000210.

<https://doi.org/10.1371/journal.pwat.0000210>

Shehabi, A., Smith, S. J., Hubbard, A., Newkirk, A., Lei, N., Siddik, M. A. B., Holecek, B., Koomey, J., Masanet, E., & Sartor, D. (2024). 2024 United States data center energy usage report (LBNL-2001637). Lawrence Berkeley National Laboratory.

<https://eta.lbl.gov/publications/2024-united-states-data-center>

Zhao, M., Wang, X., & Bayram, I. S. (2021). A synthetic approach for datacenter power consumption regulation. *Energies*, 14(9), Article 2601.

<https://doi.org/10.3390/en14092601>